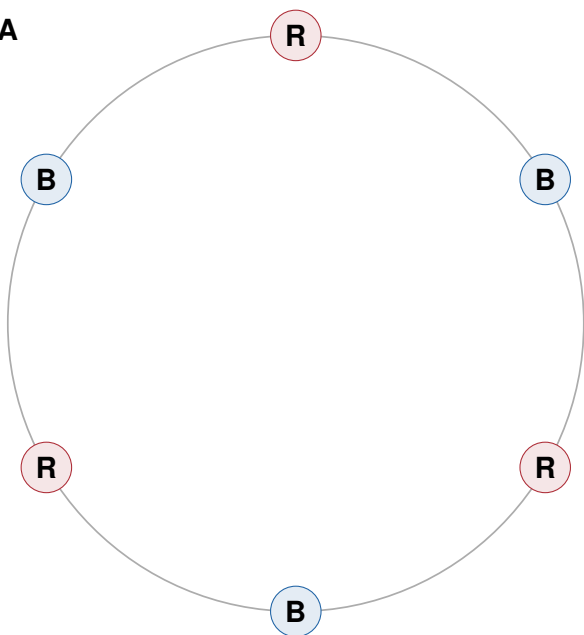


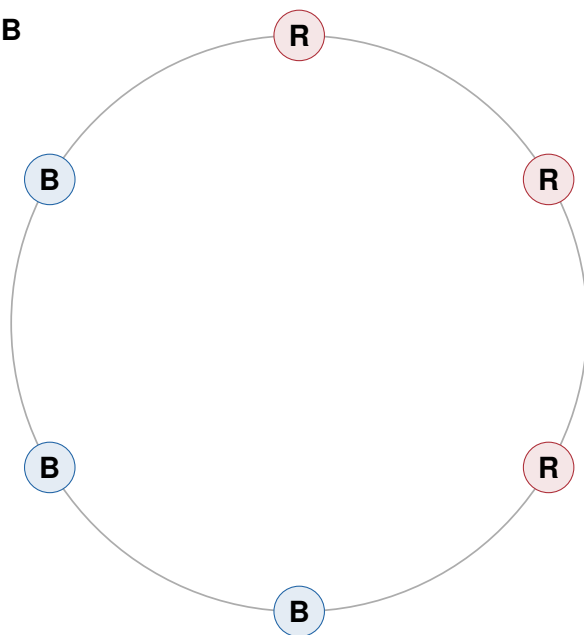
Join each R to a B, using every dot once. Lines must stay inside the circle and must not cross. The dots keep their places.

Problem 1: Find every pairing for each arrangement.

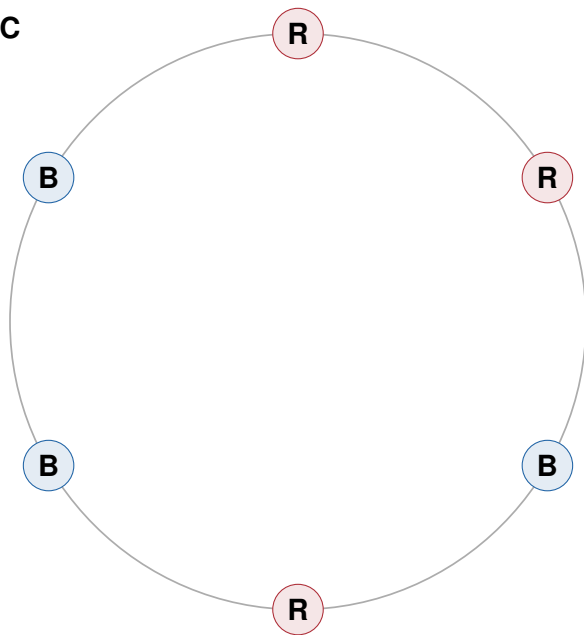
A



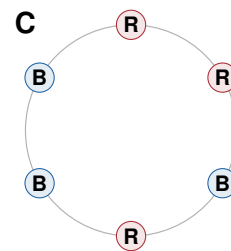
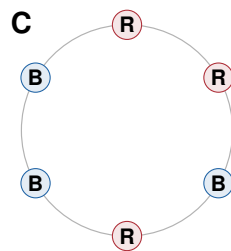
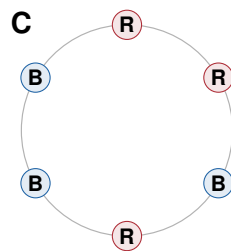
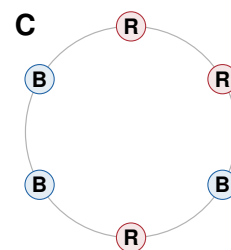
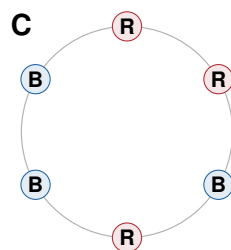
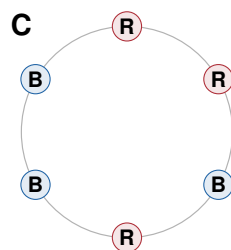
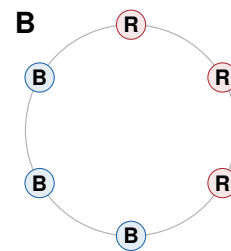
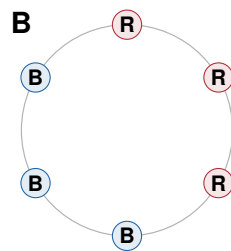
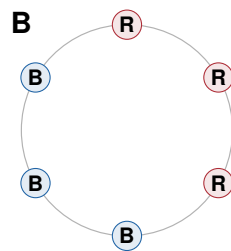
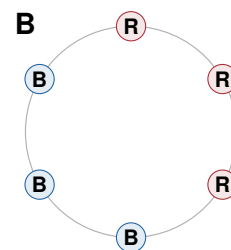
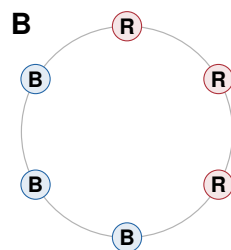
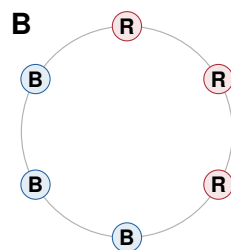
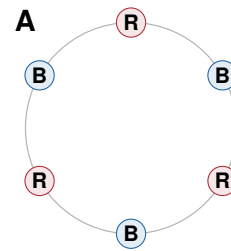
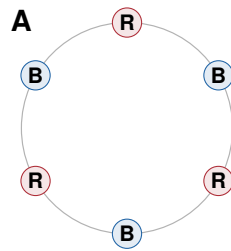
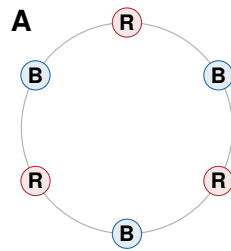
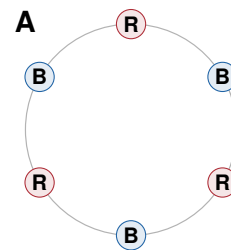
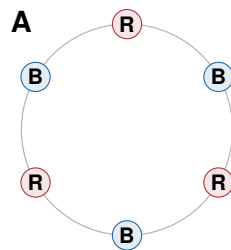
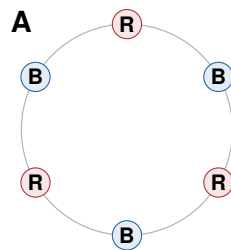
B



C

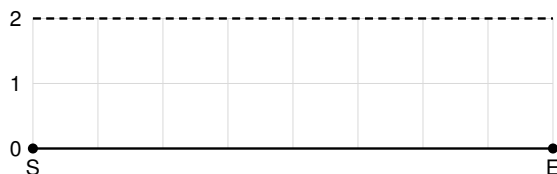
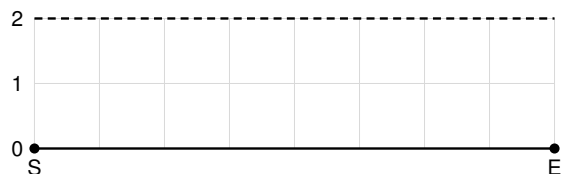
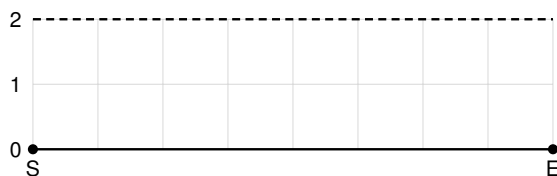
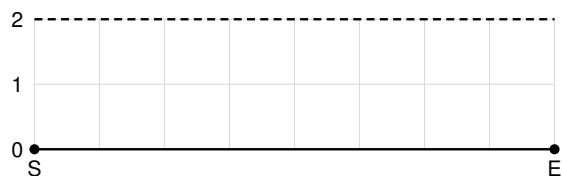
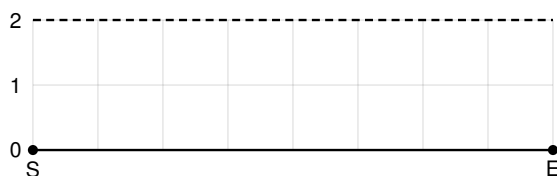
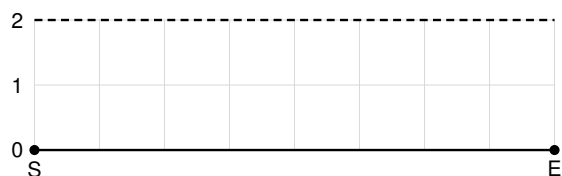
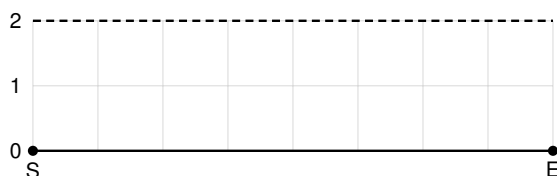
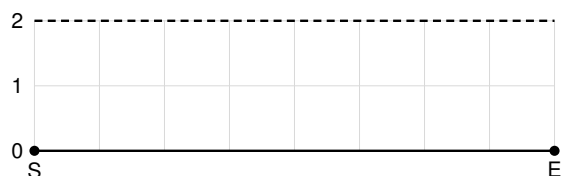
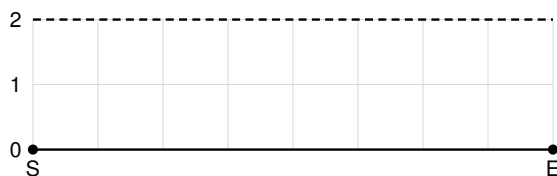
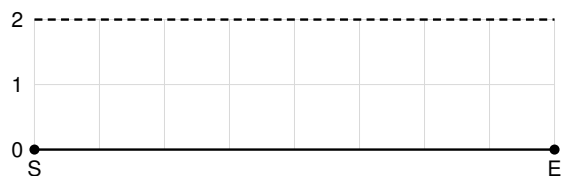


Extra workspace



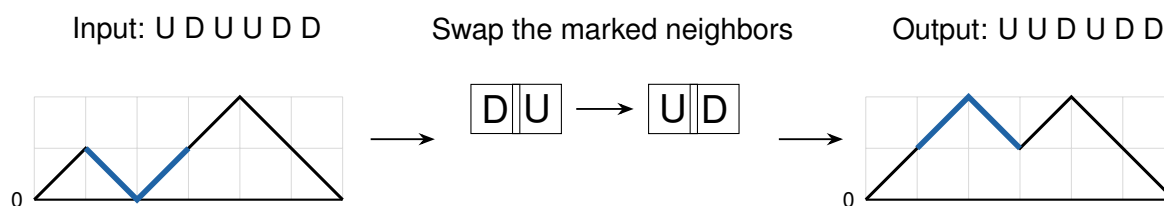
U is one diagonal step up and right. D is one diagonal step down and right. A path starts at S and ends at E; it stays on or above 0.

Problem 2: Use four U steps and four D steps. Find every path that stays at or below the dashed ceiling at height 2.

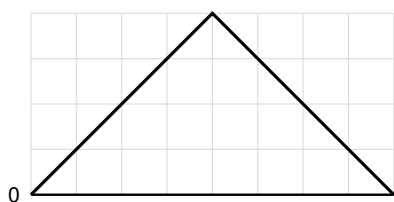


Problem 3: With the same ceiling, how many paths use five U steps and five D steps? Find a way to know without drawing every path.

A move swaps neighboring U and D cards. Stay on or above 0 after every move.

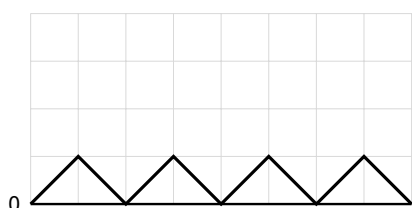


Problem 4: Show a shortest route from each start to the target. Can you tell how many moves a path needs before making any moves?

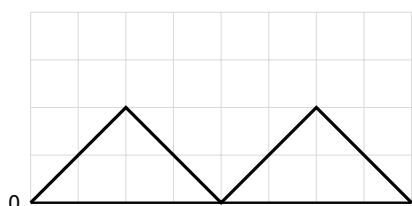


Target: U U U U D D D D

Start: U D U D U D U D



Start: U U D D U U D D



Start: U U U D U D D D

